



Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-5.
Department of Electronics and Telecommunication Engineering

Program: E&TC/ ECE
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Smart Pest Identifier Framework using Multi-model Deep Learning Approach

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Overview:

1. Introduction/Project Idea
2. Objective
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Introduction / Idea of Project

Farmers often describe pests verbally to pesticide sellers, leading to wrong pesticide recommendations. This results in multiple chemical applications, harming crop health, soil quality, and consumer safety.

Our system uses image-based pest identification to give farmers the exact pest name, enabling correct pesticide use and reducing excessive chemical application.

Farmers often face challenges in identifying pests accurately, especially when:

1. Multiple pest species co-occur.
2. Pests have similar morphological features.
3. Images are degraded or unclear.
4. New/unseen pests appear.

Objective

This project proposes a multi-model deep learning framework that identifies pests and handles complex real-world scenarios through specialized models for different problem types.

By leveraging **computer vision, transformer architectures, and ensemble learning**, the system enables real-time, accurate, and accessible pest identification via a **mobile application**.

Deep learning system for pest detection with specialized **sub-models for tricky cases** (similar species, poor image quality, unseen pests). Models run on the **cloud for real-time, scalable processing**, accessed via a simple, **farmer-friendly mobile app**.

Objective

Accurate Pest Identification

Develop a deep learning–based framework capable of identifying pests across multiple challenging scenarios.

Specialized Model Handling

Create sub-models optimized for specific issues such as co-occurrence, morphological similarity, image degradation, and unseen species.

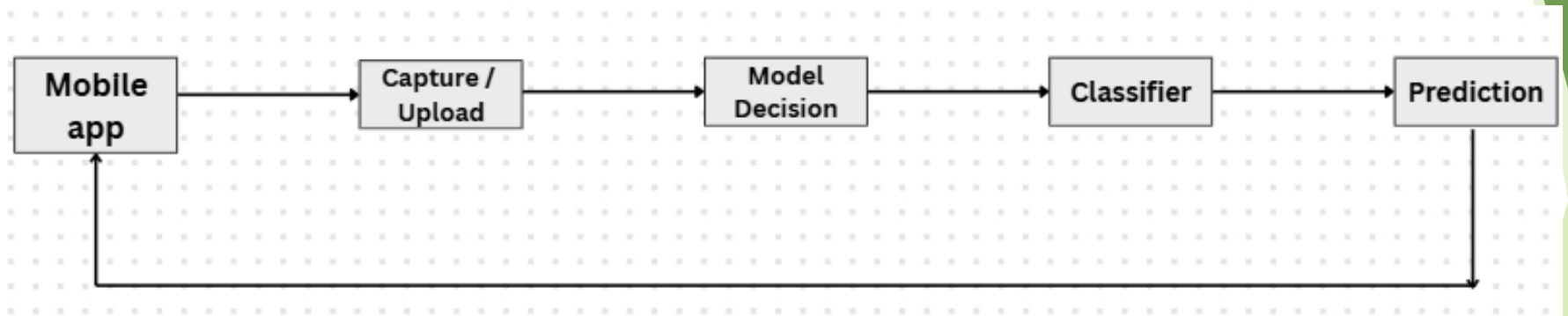
Cloud-Integrated Deployment

Host pre-trained models on cloud infrastructure to allow real-time processing and scalability.

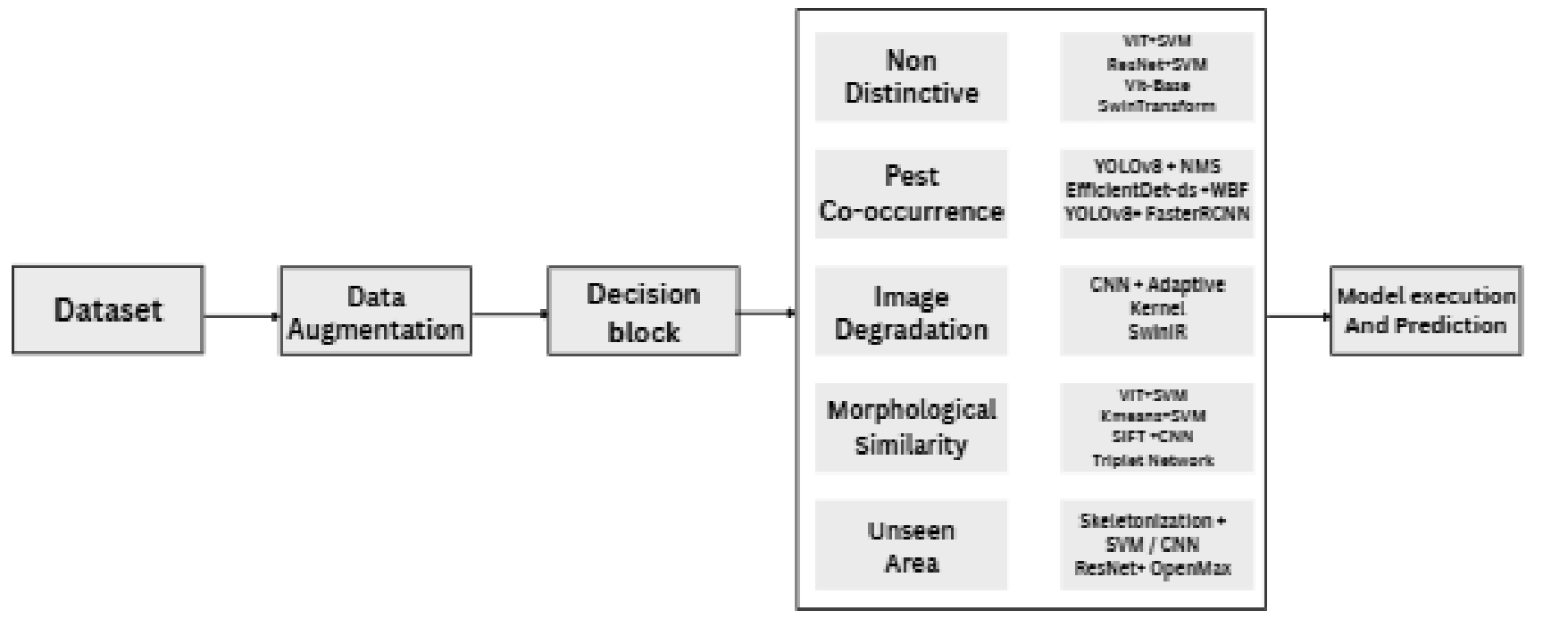
Farmer-Centric Mobile Interface

Design an intuitive mobile application for easy pest image capture, uploading, and result retrieval.

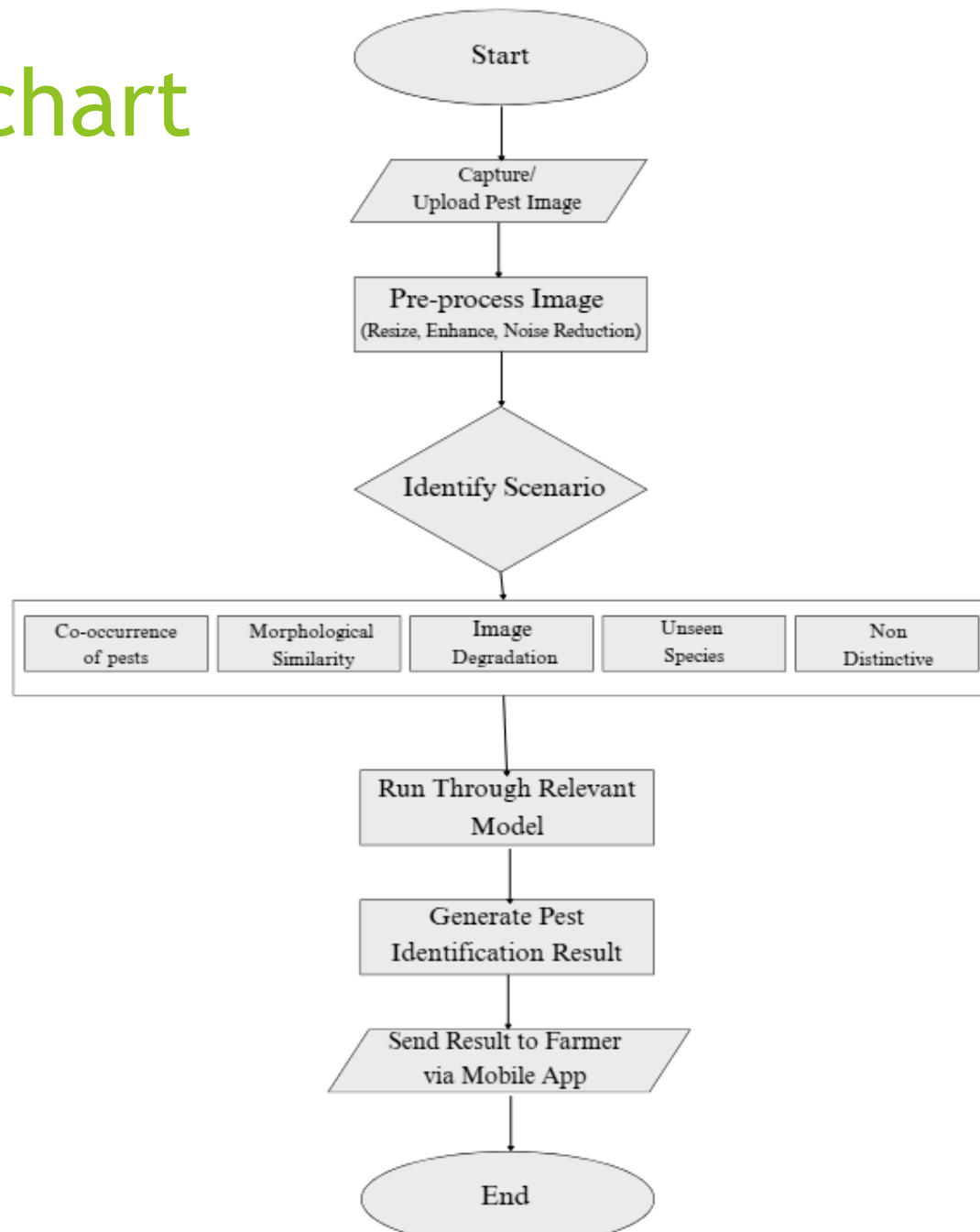
Block Diagram



Block Diagram



Flowchart



Working Flow

Step 1 - Image Acquisition

- Farmer uses the mobile app to capture a live photo or upload from gallery.

Step 2 - Preprocessing

- Standardization: Resizes, normalizes, and formats the image for model compatibility.

Step 3 - Problem Type Decision Block

- A meta-classifier quickly analyzes the image and assigns it to one of the five sub-problems:

Step 4 - Specialized Model Execution (Cloud)

- Image is routed to the corresponding pre-trained deep learning model

Working Flow

Step 5 - Result Generation

- Pest name(s) with confidence scores.

Step 6 - Mobile App Output & Feedback Loop

- Farmer gets results in simple UI.
- Farmer can confirm or correct pest detection - feedback is logged for future model improvement.

Applications

1. Multi-Pest Detection in the Same Field

- Identifies and classifies multiple pests present in a single image.

2. Differentiating Visually Similar Pests

- Prevents misidentification of look-alike pests, which often leads to wrong pesticide use.

3. Accurate Classification of Ambiguous Pests

- Detects pests that lack clear visual traits due to partial visibility or early growth stage.

4. Reliable Detection in Poor Quality Images

- Handles images affected by motion blur, low light, noise, or compression artifacts.

5. Identification of Rare or New Pest Species

- Creates opportunities for real-time pest discovery and reporting to agricultural boards.